

SWOT Analysis

	Helpful	Harmful
Internal	Strengths 1. Special economic zones (SEZs) 2. Proactive approach of government and Make in India campaign 3. Skill, labour and human capital 4. Resident expertise in semiconductor design and embedded software 5. Incentives and policies from central and state government	Weaknesses 1. Absence of SMT manufacturing firms 2. Dependence on import 3. Lack of infrastructure 4. Consumption-driven market 5. Lack of domestic standards and advanced facilities for manufacturing and testing 6. Non-availability of finance 7. Ineffective implementation of government incentives 8. Statutory obligations and licences
External	Opportunities 1. Growing electronics market 2. Penetration and booming demand for electronics in defence, lighting and automotive industry 3. Rising manufacturing cost in countries like China 4. Emphasis of DeitY on indigenous electronics manufacturing 5. Government support for infrastructure development 6. Availability of innovations and latest technologies 7. Untapped manpower in rural areas	Threats 1. Import of equipment for domestic purposes 2. High sales tax rates and duties for domestic manufacturers 3. Highly established SMT equipment market in other countries 4. Unfavourable labour laws

Different types of SMT equipment

1. Stencil/screen printer
 - Manual
 - Semi-automatic
 - Automatic
2. SMT pick-and-place equipment
 - High speed
 - Medium speed
 - Low speed
3. Soldering equipment
 - Reflow oven
 - Wave oven
4. Solvent cleaning equipment
5. Inspection equipment
 - X-ray
 - Laser
 - Optical
6. Repair and rework equipment
 - Manual
 - Semi-automatic

A report from McKinsey and Company suggests that Indian manufacturers lag behind their global peers in key operational areas, and that India's manufacturing sector has the potential to touch US\$ one trillion by 2025. To meet the ever-increasing demand of electronics goods, success in the R&D sector is not just enough.

We have to generate sources of raw materials and manufacture the designs in India. This means we need to have an SMT assembly in-house.

Currently, Indian SMT industry is largely dependent on imports for manufacturing equipment as well as raw materials for the processes. Most of the SMT equipment brands have subsidiaries, agents or distributors across the country. But proper availability of spares and services for imported capital equipment is still a challenge. However, projections of the industry, along with the electronics indigenisation movement of Department of Electronics and Information Technology (DeitY) by government of India (GOI), look extremely favourable for the SMT equipment market here.

Driving segments

The upsurge in the number of electronic products and the penetration of electronics into almost all fields of the industry are positive factors for SMT equipment businesses. In fact, SMT equipment manufacturers are modifying the instruments' capabilities for the purpose of meeting industry demands.

Lighting. One of the major growth areas for SMT equipment manufacturers is the lighting industry. With an increasing popularity of light emitting diodes (LEDs), high-speed and high-capacity manufacturing equipment for specifically handling LEDs and special components (for example, high thermal dissipative LEDs) are being introduced into the Indian market.

Automobile. Latest advances and enhancements in automotive infotainment have resulted in increasing the complexity of circuits, thus requiring SMT lines that are very precise and can withstand an extended operating temperature.

India is slightly lagging behind in the manufacture of high-technology components and manufacturing units. However, through ramping up of indigenous manufacturing in telecommunication and defence industries, and development of R&D centres focussing on SMT, the next generation of advanced electronic manufacturing equipment would be introduced.

Availability of manpower. One of the major reasons why international firms prefer India as their manufactur-